

Gradient of a line



- 1
- a The line has a gradient of zero.
 - b The line has an undefined gradient.
 - c The line has a positive gradient.
 - d The line has a negative gradient.

- 2
- a
 - i 2
 - ii 6
 - iii $\frac{1}{3}$
 - b
 - i 12
 - ii 6
 - iii 2
 - c
 - i 14
 - ii 7
 - iii 2
 - d
 - i 12
 - ii 4
 - iii 3
 - e
 - i -12
 - ii 4
 - iii -3
 - f
 - i -6
 - ii 2
 - iii -3
 - g
 - i -6
 - ii 6
 - iii -1
 - h
 - i 8
 - ii 8
 - iii 1

- 3 Because a vertical line has no run. So the formula for the gradient $\frac{\text{rise}}{\text{run}}$ will have a denominator of zero, and any fraction with a zero denominator is undefined.

4 0

- 5
- a 2
 - b $-\frac{7}{5}$
 - c $\frac{1}{2}$
 - d $-\frac{2}{3}$

Gradient formula

6 $-\frac{3}{2}$

7 No

- 8
- a -2
 - b As x increases, the value of y decreases.

- 9
- a 3
 - b $\frac{3}{5}$
 - c -4
 - d -3
 - e 0
 - f 0
 - g 1
 - h -1

10

a $\frac{5}{3}$

b $\frac{5}{3}$

c Yes



11 $m = 1$

12

a $t = 3$

b $t = 15$

c $d = 20$

d $c = -\frac{12}{7}$

13

a $Q(4, -2)$

b $R(-4, -18)$

c 2

14

a $y = 27$

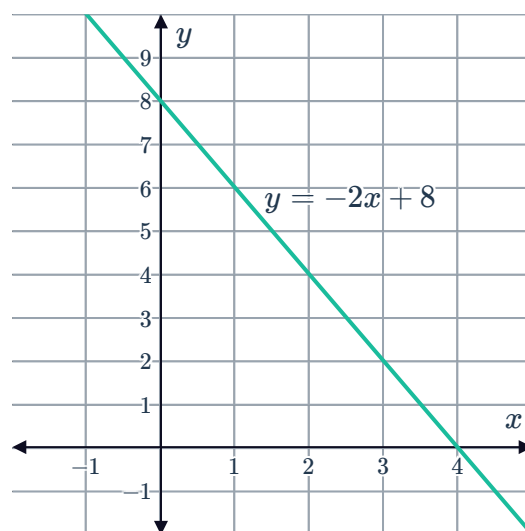
b 5

15

a i $y = 8$

ii $x = 4$

b



c -2

16

a $y = 1$

b $y = 1$

c

i 1

ii -1

d -1



Properties of polygons

17

a

i $-\frac{2}{3}$

ii $-\frac{2}{3}$

iii 1

iv 1

- b A parallelogram because opposite sides are parallel because they have the same gradient.

18

a -1

b 1

c -1

d Yes

19

a -1

b -1

c Yes

d Yes

e Parallelogram

20

a $D\left(\frac{13}{2}, -4\right)$ and $E\left(\frac{1}{2}, -\frac{17}{2}\right)$

b $\frac{3}{4}$

c $\frac{3}{4}$

d Yes

21

a BC

b 15 units²

Applications

22

a 0.9

b No

23

a -0.72

b -0.35

c Run A

24 890 m